

CAPSCAN PRO



CSE 200: Software Development Project II

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CapScan

Introduction

CapScan is a simple, reliable tool that makes sure your memory cards and USB drives are actually what they claim to be. It checks their real capacity, keeps your data safe, measures read/write speeds, and ensures everything works smoothly. With so many fake or faulty storage devices around, CapScan helps you avoid nasty surprises like lost files or corrupted data.

Key Features

Capacity Scan

Many fake or faulty storage devices show more space than they actually have. For example, a drive may claim to be 128 GB but really only store 32 GB. CapScan checks the *real usable size* and warns the user if the advertised size is false.

Data Integrity

Sometimes files get saved incorrectly or become corrupted without notice. CapScan writes test data and then reads it back to see if it matches. If there are errors, the software shows that the device may not be safe for storing important files.

Speed Test

Storage devices work at different speeds. CapScan tests how fast the device can *write* (save) and *read* (open) files. This helps users know if the device is suitable for things like large file transfers, video editing, or backups.

Technical Importance

CapScan makes life easier for many types of users:

- **Everyday Users** – Feel confident that your USB drives, memory cards, or hard drives are safe before storing family photos, important documents, or backups.
- **Students** – Keep assignments, projects, and research files secure by checking that pen drives and portable drives are working properly.
- **Shops & Retailers** – Build trust with customers by testing storage devices before selling them, avoiding the risk of fake or faulty products.
- **Schools & Colleges** – Protect valuable digital learning materials, exam results, and school records with reliable storage checks.
- **IT Professionals** – Ensure that SSDs, RAID setups, and backup drives deliver the speed and reliability needed for demanding tasks.
- **Forensic & Data Recovery Experts** – Safely verify evidence drives and prevent further damage when handling sensitive data.

Use Cases

1. Detect drives that claim more space than they really have.
2. Make sure external drives safely store your important files.
3. Check how fast SSDs perform for games, videos, or heavy tasks.
4. Verify drives used in investigations or data recovery.

5. Give students, professionals, and businesses trust in their storage.